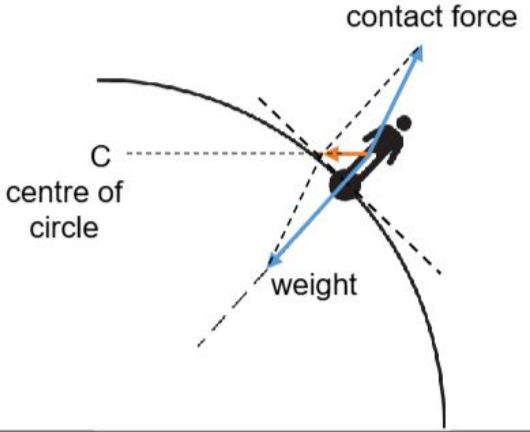
Paper 2 (75 marks)

1ai	Resultant force = centripetal force $= mr\omega^2 = m(R\cos 52.2)\left(\frac{2\pi}{T}\right)^2$ $= 58.6 \times 6.37 \times 10^6 \cos 52.2 \times \left(\frac{2\pi}{24 \times 60 \times 60}\right)^2$ = 1.21 N	C1 C1 A1
1aii	arrow from student pointing along the dotted line, labelled 'weight' upwards arrow from student pointing in a direction to the left of normal and above the tangent to the Earth, labelled 'contact force'  	B1 B1
1b	Apparent weight (which is the normal contact force) is the difference between gravitational force and the component of centripetal force along the radial direction. For same angular velocity, larger radius means larger centripetal force required at equator. Hence, apparent weight is smaller at equator.	M1 A1
2ai	Gravitational force provides centripetal force	B1